

Chapter 14

Evaluation Studies: From Controlled to Natural Settings

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Objectives

The main aims of this chapter are to:

- Explain how to do usability testing.
- Outline the basics of experimental design.
- Describe how to do field studies.



14.1 Introduction

Imagine you have designed a new shared web space intended for advertising second-hand goods. How would you find out whether householders would be able to use it to find what they wanted and whether it was a reliable and effective service? What evaluation methods would you employ?

In this chapter we describe evaluation studies that take place in a spectrum of settings, from controlled laboratories to natural settings. Within this spectrum we focus on usability testing which takes place in usability labs; experiments which take place in research labs; and field studies which take place in natural settings such as people's homes, schools, work, and leisure environments.

14.2 Usability Testing

The usability of products has traditionally been tested in controlled laboratory settings. This approach emphasizes how usable a product is. It has been most commonly used to evaluate desktop applications, such as websites, word processors, and search tools. Doing usability testing in a laboratory, or a temporarily assigned controlled environment, enables evaluators to control what users do and to control environmental and social influences that might impact the users' performance. The goal is to test whether the product being developed is usable by the intended user population to achieve the tasks for which it was designed.

14.2.1 Methods, Tasks, and Users

Collecting data about users' performance on predefined tasks is a central component of usability testing. As mentioned in [Chapter 13](#), a combination of methods is often used to collect data. The data includes video recordings of the users including their facial expressions and logged keystrokes and mouse movements. Sometimes participants are asked to think aloud while carrying out tasks, as a way of revealing what they are thinking and planning (see Section 14.2.3). In addition, a user satisfaction questionnaire is used

to find out how users actually feel about using the product, by asking them to rate it along a number of scales, after interacting with it. Structured or semi-structured interviews may also be conducted with users to collect additional information about what they liked and did not like about the product. Sometimes evaluators also collect data about how the product is used in the field.

Examples of the tasks that are given to users include searching for information, reading different typefaces (e.g. Helvetica and Times), and navigating through different menus. Performance times and numbers are the two main performance measures used, in terms of the time it takes typical users to complete a task, such as finding a website, and the number of errors that participants make, such as selecting wrong menu options when creating a spreadsheet. The quantitative performance measures that are obtained during the tests produce the following types of data (Wixon and Wilson, 1997):

- Time to complete a task.
- Time to complete a task after a specified time away from the product.
- Number and type of errors per task.
- Number of errors per unit of time.
- Number of navigations to online help or manuals.
- Number of users making a particular error.
- Number of users completing a task successfully.

In the early days of usability testing, tests were conducted to investigate specific features of an interface. For example, a team of scientists from Xerox Corporation ran a series of tests to determine what was the optimal number of buttons to put on a mouse, how many items to put in a menu, and how to design icons, as part of their Xerox Star office workstation system (Bewley *et al*, 1990). In total, over 15 tests were performed involving over 200 users, lasting over 400 hours. The results of the various tests were then fed back into the design of the interface.

A key concern is the number of users that should be involved in a usability study: five to twelve is considered an acceptable number (Dumas and Redish, 1999), but sometimes it is possible to use fewer when there are budget and schedule constraints. For instance, quick feedback about a design idea, such as the initial placement of a logo on a website, can be obtained from only two or three users.

14.2.2 Labs and Equipment

Many companies, such as Microsoft and IBM, used to test their products in custom-built usability labs (Lund, 1994). These facilities comprise a main testing laboratory, with recording equipment and the product being tested, and an observation room where the evaluators watch what is going on and analyze the data. There may also be a reception area for testers, a storage area, and a viewing room for observers. The space may be arranged to superficially mimic features of the real world. For example, if the product is an office product or for use in a hotel reception area, the laboratory can be set up to look like those environments. Soundproofing and lack of windows, telephones, co-workers, and other workplace and social artifacts eliminate most of the normal sources of distraction so that the users can concentrate on the tasks set for them to perform.

Typically there are two to three wall-mounted video cameras that record the users' behavior, such as hand movements, facial expression, and general body language. Microphones are also placed near where the participants will be sitting to record their utterances. Video and other data is fed through to monitors in the observation room. The observation room is usually separated from the main laboratory or workroom by a one-way mirror so that evaluators can watch participants being tested but testers cannot see them. It can be a small auditorium with rows of seats at different levels or, more simply, a small backroom consisting of a row of chairs facing the monitors. They are designed so that evaluators and others can watch the tests while ongoing, both on the monitors and through the mirror. [Figure 14.1](#) shows a typical arrangement.



Figure 14.1 A usability laboratory in which evaluators watch participants on a monitor and through a one-way mirror

Sometimes, modifications may have to be made to the room set up to test different types of applications. For example, Nodder and his colleagues at Microsoft had to partition the space into two rooms when they were testing NetMeeting, an early videoconferencing product, in the mid-1990s, as [Figure 14.2](#) shows (Nodder *et al*, 1999). This allowed users in both rooms to be observed when conducting a meeting via the videoconference system.

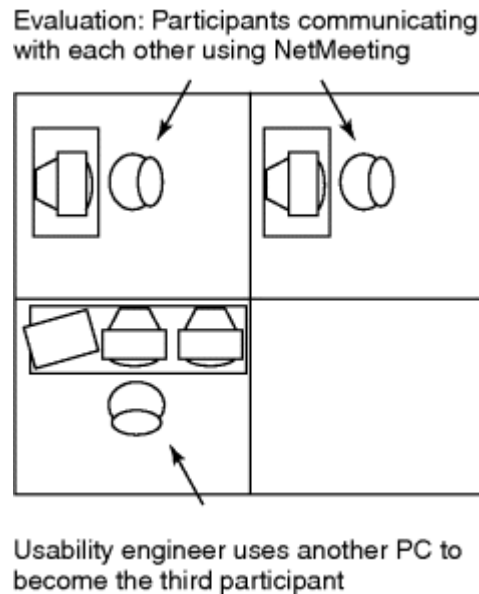


Figure 14.2 The testing arrangement used for NetMeeting, an early videoconferencing product

Usability labs can be very expensive and labor-intensive to run and maintain. A less expensive alternative, that started to become more popular in the early and mid-90s, is the use of mobile usability testing equipment. Video cameras, laptops, and other measuring equipment are temporarily set up in an office or other space, converting it into a makeshift usability laboratory. Another advantage is that equipment can be taken into work settings, enabling testing to be done on site, making it less artificial and more convenient for the participants.

There is an increasing number of products that are specifically designed for mobile evaluation. Some are referred to as lab-in-a-box or lab-in-a-suitcase because they pack away neatly into a convenient carrying case. One example is the Tracksys portable lab that costs under \$5000 depending on exactly what equipment is included ([Figure 14.3](#)). It is composed of off-the-shelf components that plug into a PC and can record video direct to hard disk ([Figure 14.4](#)). Some recent additions to Tracksys include: a system called the GoToMeeting package, an easier remote control system including a remote eye-tracker, and a mobile head-mounted eye-tracker. The latter is fitted to a helmet or lightweight cap so that the system offers complete portability for research in sports, usability studies, and a

variety of other disciplines ([Figure 14.5](#)). There are also mobile eye-tracking and eye-gaze systems available that can be used.



Figure 14.3 The Tracksys lab-in-a-box system, which comprises components that pack into a heavy duty padded flight case plus a PC system

Source: Courtesy of Harry Brignull.



Figure 14.4 The Tracksys system being used with a mobile device camera that attaches to a flexible arm, which mounts on a mobile device, and is tethered to the lab

Source: Courtesy of Harry Brignull.



Figure 14.5 The mobile head-mounted eye-tracker

Source: Picture courtesy of SensoMotoric Instruments (SMI), copyright 2010.

Another trend has been to conduct remote usability testing, where users perform a set of tasks with a product in their own setting and their interactions with the software are logged remotely. An advantage of this approach is that many users can be tested at the same time and the logged data automatically compiled into statistical packages for data analysis. For example, the number of clicks per

page and the tracking of clicks when searching websites for specified tasks can be readily obtained.

14.2.3 An Example of Usability Testing: The iPad

Usability specialists Budiu and Nielsen (2010) from the Nielsen Norman Group conducted a usability test of the websites and apps specific to the iPad. They wanted to understand how the interactions with the device affected people and to get feedback to their clients and developers as well as those who were eager to know if the iPad lived up to the hype – which was being reported at the time it came to market. They used two methods: usability testing with think-aloud in which users said what they were doing and thinking as they did it, and an expert review. Here we describe the usability testing they conducted. A key question they asked was: ‘Are user expectations different for the iPad compared with the iPhone?’ A previous study of the iPhone showed people preferred using apps than browsing the web because the latter was slow and cumbersome at that time. Would this be the same for the iPad, where the screen was larger and web pages were more similar to how they appeared on the laptops or desktop computers that most people were used to using? Budiu and Nielsen also hoped that their study would be able to address the question that many companies were considering at that time: whether it was worth developing specific websites for the iPad (in the way some companies were doing for smartphones) or would the desktop versions be acceptable when interacted with using the iPad multitouch interface.

The usability testing was carried out in two cities in the United States: Fremont, California, and Chicago. The test sessions were similar; the aim of both was to understand the typical usability issues that people encounter when using applications and accessing websites on the iPad. Seven participants were recruited: all were experienced iPhone users who had owned their phones for at least 3 months and who had used a variety of apps. One participant was also an iPad owner. One reason for selecting participants who used iPhones was because they would have previous experience of using apps and the web with a similar interaction style to the iPad.

The participants were considered typical users. They varied in age and occupation. Two participants were in their 20s, three were in their 30s, one in their 50s, and one in their 60s. Their occupations were: food server, paralegal, medical assistant, retired food driver, account rep, and homemaker. Three were males and four were females.

Before taking part, the participants were asked to read and sign an informed consent form agreeing to the terms and conditions of the study. This form described:

- what the participant would be asked to do;
- the length of time needed for the study;
- the compensation that would be offered for participating;
- the participants' right to withdraw from the study at any time;
- a promise that the person's identity would not be disclosed; and
- an agreement that the data collected would be confidential and would not be made available to marketers or anyone other than the evaluators.

The Tests

The session started with participants being invited to explore any application they found interesting on the iPad. They were asked to comment on what they were looking for or reading, what they liked and disliked about a site, and what made it easy or difficult to carry out a task. A moderator sat next to each participant and observed and took notes. The sessions were video recorded and lasted about 90 minutes. Participants worked on their own.

After exploring the iPad they were asked by the evaluator to open specific apps or websites and to explore them, then carry out one or more tasks as they would if they were on their own. Each was assigned the tasks in a randomized order. All the apps that were tested were designed specifically for the iPad, but for some tasks the users were asked to do the same task on a website that was not specifically designed for the iPad. For these tasks the evaluators

took care to balance the presentation order so that the app would be first for some participants and the website would be first for others. Over 60 tasks were chosen from over 32 different sites. Examples are shown in [Table 14.1](#).

TABLE 14.1

**Examples of some of the tests used in the iPad evaluation
(adapted from Budiu and Nielsen, 2010).**

Source: Copyright Nielsen Norman Group, from report available at <http://www.nngroup.com/reports/>.

App or website	Task
iBook	Download a free copy of <i>Alice's Adventures in Wonderland</i> and read through the first few pages.
Craigslist	Find some free mulch for your garden.
eBay	You want to buy a new iPad on eBay. Find one that you could buy from a reputable seller.
<i>Time</i> Magazine	Browse through the magazine and find the best pictures of the week.
Epicurious	You want to make an apple pie for tonight. Find a recipe and see what you need to buy in order to prepare it.
Kayak	You are planning a trip to Death Valley in May this year. Find a hotel located in the park or close to the park.

Activity 14.1

1. What was the main purpose of this study?
2. What aspects are considered to be important for good usability and user experience?
3. How representative are the tasks outlined in [Table 14.1](#) of what a typical iPad user might do?

Comment

Show/Hide

The Equipment

The testing was done using a setup similar to the mobile usability kit shown in [Figure 14.6](#). A camera recorded the participant's interactions and gestures when using the iPad and streamed the recording to a laptop computer. A webcam was also used to record the expressions on the participants' faces and their think-aloud commentary. The laptop ran software called Morae, which synchronized these two data streams. Up to three observers (including the moderator sitting next to the participant) watched the video streams (rather than observing the participant directly) on their laptops situated on the table – meaning they did not have to invade the participants' personal space.



Figure 14.6 The setup used in the Chicago usability testing sessions

Source: Copyright Nielsen Norman Group, from report available at <http://www.nngroup.com/reports/>.

Usability Problems

The main findings from the study showed that the participants were able to interact with websites on the iPad but that it was not optimal. For example, links on the pages were often too small to tap on reliably and the fonts were sometimes difficult to read. The various usability problems identified in the study were classified according to a number of well-known interaction design principles and concepts, including: mental models, navigation, the quality of images, problems of using a touch screen with small target areas, lack of affordances, getting lost in the application, the effects of changing orientations, working memory, and the feedback that they received.

An example of a navigation problem identified during the evaluation was that when the cover of *Time* magazine appeared on the iPad, it didn't contain any hyperlinks. The contents page did but it was not

easily accessible. In order to access it, users first had to tap the screen to reveal the controls at the bottom, then select the Contents button to display the contents carousel, and then select the contents page in the carousel.

Another problem they identified was that the participants sometimes did not know where to tap on the iPad to select options such as buttons and menus. It should be obvious from the affordances of an interface as to where they are and how to select them. However, the evaluators found that in many cases the participants repeatedly tapped the iPad interface in order to initiate an action, such as when trying to select an option from a menu.

Getting lost in an application is an old but important problem for designers of digital products and some participants got lost because they tapped the iPad too much and could not find a back button and could not get themselves back to the home page. One participant said “ . . . I like having everything there [on the home page]. That's just how my brain works” (Budiou and Nielsen, 2010, p. 58). Other problems arose because applications appeared differently in the two views possible on the iPad: portrait and landscape. More information was provided by some app developers in the horizontal (landscape) view than in the vertical (portrait) view, which caused problems for some participants.

Interpreting and Presenting the Data

Based on the findings of their study, Budiou and Nielsen made a number of recommendations, including supporting standard navigation. The results of the study were written up as a report that was made publically available to app developers and the general public (it is available from www.nngroup.com). It provided a summary of key findings for the general public as well as specific details of the problems the participants had with the iPad, so that developers could decide whether to make specific websites and apps for the iPad.

While being revealing about how usable websites and apps are on the iPad, the usability testing was not able to reveal how it will be used in people's everyday lives. This would require an in the wild

study where observations are made of how people use them in their own homes and when traveling.

Activity 14.2

1. Was the selection of participants for the iPad study appropriate? Justify your comments.
2. What might have been the problems with asking participants to think out loud as they completed the tasks?

Comments

Show/Hide

Video produced in 2010 by Blue Duck Labs illustrates some special considerations that evaluators need to know and include when testing systems with young children. It describes having a young child test a paper prototype of a potential coloring program for kids – see it at <http://youtu.be/9wQkLthhHKA>

When more rigorous testing is needed, a set of standards can be used for guidance – such an approach is described in Box 14.1 for the development of the US Government's Recovery.gov website. For this large website, several methods were used including: usability testing, expert reviews (discussed in [Chapter 15](#)), and focus groups.

BOX 14.1

Ensuring Accessibility and Section 508 Compliance for the Recovery.gov Website (Lazar et al, 2010b)

The American Recovery and Reinvestment Act (informally known as 'the Stimulus Bill') became law on February 17, 2009, with the intention of infusing \$787 billion into the US economy to create jobs and improve economic conditions. The Act established an independent board, the Recovery Accountability and Transparency Board, to oversee the spending and detect, mitigate, and minimize any waste, fraud, or abuse. The law required the Board to establish a website to provide the public with information on the progress of the recovery effort. A simple website was launched the day that the Act was signed into law, but one of the immediate goals of the Board was to create a more detailed website, with data, geospatial features, and Web 2.0 functionality, including data on every contract related to the Act. The goal was to provide political transparency at a scale not seen before in the US federal government so that citizens could see how money was being spent.

A major goal in the development of the Recovery.gov website was meeting the requirement that it be accessible to those with disabilities, such as perceptual (visual, hearing) and motor impairments. It had to comply with guidelines specified in Section 508 of the Rehabilitation Act (see the id-book.com website for details). At a broad level, three main approaches were used to ensure compliance:

- Usability testing with individual users, including those with perceptual and motor impairments.
- Routine testing for compliance with Section 508 of the Rehabilitation Act, done every 3 months, using screenreaders such as JAWS, and automated testing tools such as Watchfire.

- Providing an online feedback loop, listening to users, and rapidly responding to accessibility problems.

During development, ten 2-hour focus groups with users were convened in five cities. An expert panel was also convened with four interface design experts, and usability testing was performed, specifically involving 11 users with various impairments. Several weeks before the launch of [Recovery.gov](https://www.recovery.gov) 2.0, the development team visited the Department of Defense Computer Accommodations Technology Evaluation Center (CAPTEC) to get hands-on experience with various assistive technology devices (such as head-pointing devices) which otherwise would not be available to the Recovery Accountability and Transparency Board in their own offices.

Approaches were developed to meet each compliance standard, including situations where existing regulations don't provide clear guidance, such as with PDF files. A large number of PDF files are posted each month on [Recovery.gov](https://www.recovery.gov), and those files also undergo Section 508 compliance testing. The files undergo automated accessibility inspections using Adobe PDF accessibility tools, and if there are minor clarifications needed, the [recovery.gov](https://www.recovery.gov) web managers make the changes; but if major changes are needed, the PDF file is returned to the agency generating the PDF file, along with the Adobe-generated accessibility report. The PDF file is not posted until it passes the Adobe automated accessibility evaluation. Furthermore, no new modules or features are added to the [Recovery.gov](https://www.recovery.gov) site until they have been evaluated for accessibility using both expert evaluations and automated evaluations. Because of the large number of visitors to the [Recovery.gov](https://www.recovery.gov) website (an estimated 1.5 million monthly visitors), ensuring accessible, usable interfaces is a high priority. The full case study is available on www.id-book.com. Since the original Act was passed, it has since been amended in 2013 so that it now includes aid victims of Hurricane Sandy, a particularly devastating hurricane that destroyed

thousands of homes and businesses along the north east coast of the USA. ■

14.3 Conducting Experiments

In research contexts, specific hypotheses are tested that make a prediction about the way users will perform with an interface. The benefits are more rigor and confidence that one interface feature is easier to understand or faster to use than another. An example of a hypothesis is: context menus (i.e. menus that provide options related to the context determined by the users' previous choices) are an easier to select option compared with cascading menus. Hypotheses are often based on a theory, such as Fitts' Law (see next chapter), or previous research findings. Specific measurements provide a way of testing the hypothesis. In the above example, the accuracy of selecting menu options could be compared by counting the number of errors made by participants for each menu type.

Hypotheses Testing

Typically, a hypothesis involves examining a relationship between two things, called variables. Variables can be independent or dependent. An independent variable is what the investigator manipulates (i.e. selects), and in the above example it is the different menu types. The other variable is called the dependent variable, and, in our example, this is the time taken to select an option. It is a measure of the user performance and, if our hypothesis is correct, will vary depending on the different types of menu.

When setting up a hypothesis to test the effect of the independent variable(s) on the dependent variable, it is usual to derive a null hypothesis and an alternative one. The null hypothesis in our example would state that there is no difference in the time it takes users to find items (i.e. selection time) between context and cascading menus. The alternative hypothesis would state that there is a difference between the two on selection time. When a difference is specified but not what it will be, it is called a two-tailed hypothesis.

This is because it can be interpreted in two ways: either the context menu or the cascading menu is faster to select options from.

Alternatively, the hypothesis can be stated in terms of one effect.

This is called a one-tailed hypothesis and would state that context menus are faster to select items from, or vice versa. A one-tailed hypothesis would be made if there was a strong reason to believe it to be the case. A two-tailed hypothesis would be chosen if there was no reason or theory that could be used to support the case that the predicted effect would go one way or the other.

You might ask why you need a null hypothesis, since it seems the opposite of what the experimenter wants to find out. It is put forward so that the data can reject a statement without necessarily supporting the opposite statement. If the experimental data shows a big difference between selection times for the two menu types, then the null hypothesis that menu type has no effect on selection time can be rejected, which is different from saying that there is an effect. Conversely, if there is no difference between the two, then the null hypothesis cannot be rejected (i.e. the claim that context menus are faster to select options from is not supported).

In order to test a hypothesis, the experimenter has to set up the conditions and find ways to keep other variables constant, to prevent them from influencing the findings. This is called the experimental design. Examples of other variables that need to be kept constant for both types of menus might include size and screen resolution. For example, if the text is in 10 pt font size in one condition and 14 pt font size in the other then it could be this difference that causes the effect (i.e. differences in selection speed are due to font size). More than one condition can also be compared with the control, for example:

Condition 1 = Context menu

Condition 2 = Cascading menu

Condition 3 = Scrolling

Sometimes an experimenter might want to investigate the relationship between two independent variables: for example, age

and educational background. A hypothesis might be that young people are faster at searching on the web than older people and that those with a scientific background are more effective at searching on the web. An experiment would be set up to measure the time it takes to complete the task and the number of searches carried out. The analysis of the data would focus on both the effects of the main variables (age and background) and also look for any interactions among them.

Hypothesis testing can also be extended to include even more variables, but it makes the experimental design more complex. An example is testing the effects of age and educational background on user performance for two methods of web searching: one using a search engine and the other a browser. Again, the goal is to test the effects of the main variables (age, educational background, web searching method) and to look for any interactions among them. However, as the number of variables increases in an experimental design, it makes it more difficult to work out from the data what is causing the results.

Experimental Design

A concern in experimental design is to determine which participants to involve for which conditions in an experiment. The experience of participating in one condition will affect the performance of those participants if asked to participate in another condition. For example, having learned about the way the heart works using multimedia, if one group of participants was exposed to the same learning material via another medium, e.g. virtual reality, and another group of participants was not, the participants who had the additional exposure to the material would have an unfair advantage. Furthermore, it would create bias if the participants in one condition within the same experiment had seen the content and others had not. The reason for this is that those who had the additional exposure to the content would have had more time to learn about the topic and this would increase their chances of answering more questions correctly. In some experimental designs, however, it is

possible to use the same participants for all conditions without letting such training effects bias the results.

The names given for the different designs are: different-participant design, same-participant design, and matched-pairs design. In different-participant design, a single group of participants is allocated randomly to each of the experimental conditions, so that different participants perform in different conditions. Another term used for this experimental design is between-subjects design. An advantage is that there are no ordering or training effects caused by the influence of participants' experience of one set of tasks on their performance in the next, as each participant only ever performs in one condition. A disadvantage is that large numbers of participants are needed so that the effect of any individual differences among participants, such as differences in experience and expertise, is minimized. Randomly allocating the participants and pre-testing to identify any participants that differ strongly from the others can help.

In same-participant design (also called within subjects design), all participants perform in all conditions so only half the number of participants is needed; the main reason for this design is to lessen the impact of individual differences and to see how performance varies across conditions for each participant. It is important to ensure that the order in which participants perform tasks for this set up does not bias the results. For example, if there are two tasks, A and B, half the participants should do task A followed by task B and the other half should do task B followed by task A. This is known as counterbalancing. Counterbalancing neutralizes possible unfair effects of learning from the first task, known as the order effect.

In matched-participant design (also known as pair-wise design), participants are matched in pairs based on certain user characteristics such as expertise and gender. Each pair is then randomly allocated to each experimental condition. A problem with this arrangement is that other important variables that have not been taken into account may influence the results. For example, experience in using the web could influence the results of tests to evaluate the navigability of a website. So web expertise would be a good criterion for matching participants. The advantages and

disadvantages of using different experimental designs are summarized in [Table 14.2](#).

TABLE 14.2		
The advantages and disadvantages of different allocations of participants to conditions		
Design	Advantages	Disadvantages
Different participants	No order effects	Many participants needed. Individual differences among participants are a problem. Can be offset to some extent by randomly assigning to groups
Same participants	Eliminates individual differences between experimental conditions	Need to counterbalance to avoid ordering effects
Matched participants	Same as different participants, but the effects of individual differences are reduced	Can never be sure that subjects are matched across variables that might affect performance

The data collected to measure user performance on the tasks set in an experiment usually includes response times for subtasks, total times to complete a task, and number of errors per task. Analyzing the data involves comparing the performance data obtained across the different conditions. The response times, errors, etc. are averaged across conditions to see if there are any marked differences. Statistical tests are then used, such as *t*-tests that statistically compare the differences between the conditions, to reveal if these are significant. For example, a *t*-test will reveal whether context or cascading menus are faster to select options from.

Statistics: *t*-tests

There are many types of statistics that can be used to test the probability of a result occurring by chance but *t*-tests are the most widely used statistical test in HCI and related fields, such as psychology. The scores, e.g. time taken for each participant to select items from a menu in each condition (i.e. context and cascading menus), are used to compute the means ($\bar{x}$) and standard deviations (SDs). The standard deviation is a statistical measure of the spread or variability around the mean. The *t*-test uses a simple equation to test the significance of the difference between the means for the two conditions. If they are significantly different from each other, we can reject the null hypothesis and in so doing infer that the alternative hypothesis holds. A typical *t*-test result that compared menu selection times for two groups with 9 and 12 participants each might be: $t = 4.53$, $p < 0.05$, $df = 19$. The *t*-value of 4.53 is the score derived from applying the *t*-test; *df* stands for degrees of freedom, which represents the number of values in the conditions that are free to vary. This is a complex concept that we will not explain here other than to mention how it is derived and that it is always written as part of the result of a *t*-test. The *df*s are calculated by summing the number of participants in one condition minus one and the number of participants in the other condition minus one. It is calculated as $df = (N_a - 1) + (N_b - 1)$, where N_a is the number of participants in one condition and N_b is the number of participants in the other condition. In our example, $df = (9 - 1) + (12 - 1) = 19$. p is the probability that the effect found did not occur by chance. So, when $p < 0.05$, it means that the effect found is probably not due to chance and that there is only a 5% chance that it could be by chance. In other words there most likely is a difference between the two conditions. Typically, a value of $p < 0.05$ is considered good enough to reject the null hypothesis, although lower levels of p are more convincing, e.g. $p < 0.01$ where the effect found is even less likely to be due to chance, there being only a 1% chance of that being the case.

14.4 Field Studies

Increasingly, more evaluation studies are being done in natural settings with either little or no control imposed on participants' activities. This change is largely a response to technologies being developed for use outside office settings. For example, mobile, ambient, and other technologies are now available for use in the home, outdoors, and in public places. Typically, field studies are conducted to evaluate these user experiences.

As mentioned in [Chapter 13](#), evaluations conducted in natural settings are very different from those in controlled environments, where tasks are set and completed in an orderly way. In contrast, studies in natural settings tend to be messy in the sense that activities often overlap and are constantly interrupted. This follows the way people interact with products in their everyday messy worlds, which is generally different from how they perform on set tasks in a laboratory setting. By evaluating how people think about, interact, and integrate products within the settings they will ultimately be used in, we can get a better sense of how successful the products will be in the real world. The trade-off is that we cannot test specific hypotheses about an interface nor account, with the same degree of certainty, for how people react to or use a product – as we can do in controlled settings like laboratories. This makes it more difficult to determine what causes a particular type of behavior or what is problematic about the usability of a product. Instead, qualitative accounts and descriptions of people's behavior and activities are obtained that reveal how they used the product and reacted to its design.

Field studies can range in time from just a few minutes to a period of several months or even years. Data is collected primarily by observing and interviewing people; collecting video, audio, and field notes to record what occurs in the chosen setting. In addition, participants may be asked to fill out paper-based or electronic diaries, which run on cell phones or other handheld devices, at particular points during the day, such as when they are interrupted during their ongoing activity or when they encounter a problem when

interacting with a product or when they are in a particular location ([Figure 14.7](#)). This technique is based on the experience sampling method (ESM) discussed in [Chapter 7](#), which is often used in healthcare (Csikszentmihalyi and Larson, 1987). Data on the frequency and patterns of certain daily activities, such as the monitoring of eating and drinking habits, or social interactions like phone and face-to-face conversations, are recorded. Software running on the smartphones triggers messages to study participants at certain intervals, requesting them to answer questions or fill out dynamic forms and checklists. These might include recording what they are doing, what they are feeling like at a particular time, where they are, or how many conversations they have had in the last hour.



[Figure 14.7](#) An example of a context-aware experience sampling tool running on a mobile device

Source: From Cogdill, K. (1999) "MedlinePlus Interface Evaluation: Final Report". Reproduced by permission of Prof. Keith Cogdill.

When conducting a field study, deciding whether to tell the people being observed, or asked to record information, that they are being

studied and how long the study or session will take is more difficult than in a laboratory situation. For example, when studying people's interactions with an ambient display, telling them that they are part of a study will likely change the way they behave. Similarly, if people are using an online street map while walking in a city, their interactions may only take a few seconds and so informing them that they are being studied would disrupt their behavior.

It is important to ensure the privacy of participants in field studies. For example, photographs should not be included in reports without the participant's permission. Participants in field studies that run over a period of weeks or months, such as an investigation into how they use a type of remote control device in their homes, should be informed about the study and asked to sign an informed consent form in the usual way as mentioned in [Chapter 13](#). In situations like this, the evaluators will need to work out and agree with the participants what part of the activity is to be recorded and how. For example, if the evaluators want to set up cameras, they need to be situated unobtrusively and participants need to be informed in advance about where the cameras will be and when they will be recording their activities. The evaluators will also need to work out in advance what to do if the prototype or product breaks down. Can the participants be instructed to fix the problem themselves or will the evaluators need to be called in? Security arrangements will also need to be made if expensive or precious equipment is being evaluated in a public place. Other practical issues may also need to be taken into account depending on the location, product being evaluated, and the participants in the study.

The system for helping skiers to improve their performance (discussed in [Chapter 13](#)) was evaluated with skiers on the mountains to see how they used it and whether they thought it really did help them to improve their ski performance. A wide range of other studies have explored how new technologies have been appropriated by people in their own cultures and settings. By appropriated we mean how the participants use, integrate, and adapt the technology to suit their needs, desires, and ways of living. For example, the Opinionizer system mentioned in [Chapter 4](#) was

designed as part of a social space where people could share their opinions visually and anonymously, via a public display. The Opinionizer was intended to encourage and facilitate conversations with strangers at a party or other social gatherings. Observations of it being interacted with at a number of parties showed a honey-pot effect: as the number of people in the immediate vicinity of the system increased, a sociable buzz was created, where a variety of conversations were started between the strangers. The findings from these and other studies in natural settings are typically reported in the form of vignettes, excerpts, critical incidents, patterns, and narratives to show how the products are being appropriated and integrated into their surroundings.

14.4.1 In the Wild Studies

For several years now it has become increasingly popular to conduct in the wild studies to determine how people use and persist in using a range of new technologies or prototypes in situ. The case study in [Chapter 13](#) (Section 13.4.2) is an example of an in the wild study in which the researchers wanted to gather feedback about how a new system designed to help skiers improve their performance actually did help them and whether the skiers liked it. As mentioned previously, the term ‘in the wild’ reflects the context of the study – where the new technologies are created, deployed, and evaluated in situ (Rogers, 2011). The evaluation takes place in a natural setting in which the researchers release their prototype technology and observe from a distance how it is approached and used by people in their own homes, and elsewhere. Instead of developing solutions that fit in with existing practices, researchers often explore new technological possibilities that can change and even disrupt behavior. Opportunities are created, interventions installed, and different ways of behaving are encouraged. A key concern is how people react, change, and integrate the technology into their everyday lives. The outcome of conducting in the wild studies for different periods and at different intervals can be most revealing – demonstrating quite different results from those arising out of lab studies. Comparisons of findings from lab studies and in the wild studies have revealed that while many usability issues can be

uncovered in a lab study, actual use aspects are more difficult to discern. These aspects include how users approach the new technology, the kinds of benefits that they could derive from it, how they appropriate it in the context of everyday activity, and its sustained use over time (Rogers *et al*, 2013; Kjeldskov and Skov, 2014).

An In the Wild Study: The UbiFit Garden

Another early example of an in the wild study was the UbiFit Garden project (Consolvo *et al*, 2008), which evaluated activity sensing in the wild to address the growing problem of people's sedentary lifestyles. A mobile device was designed to encourage physical activity that used on-body sensing and activity inferencing. The personal mobile display had three components: a fitness device that monitored different types of fitness activities, an interactive application that kept detailed information, including a journal of the individual's activities, and a glanceable display that ran on cell phones. The system worked by inferring what the wearer was doing, in terms of walking, running, cycling, and using gym equipment based on data detected from accelerometer and barometer sensors. The sensor data was processed and then communicated to the cell phone using Bluetooth. The data was analyzed and used as input for the glanceable display that depicted the UbiFit Garden (see [Figure 14.8](#)). The display depicted a garden that bloomed throughout the week as the user carried out the various physical activities. A healthy regime of physical exercise was indicated by a healthy garden full of flowers and butterflies. Conversely, an unhealthy garden with not much growth or butterflies indicated an unhealthy lifestyle.



Figure 14.8 UbiFit Garden's glanceable display: (a) at the beginning of the week (small butterflies indicate recent goal attainments; the absence of flowers means no activity this week); (b) a garden with workout variety; (c) the display on a mobile phone (the large butterfly indicates this week's goal was met)

Source: From Consolvo, S., McDonald, D.W., Toscos, T. et al (2008) "Activity sensing in the wild: a field trial of UbiFit garden". In: *Proceedings of CHI 2008*, ACM Press, New York, p. 1799.

Data collection and participants.

Two evaluation methods were used in this study: an in the wild field study over 3 weeks with 12 participants and a survey with 75 respondents from 13 states across the USA that covered a range of attitudes and behaviors with mobile devices and physical activity. The goals of the in the wild study were to identify usability problems and to see how this technology fitted into the everyday lives of the six men and six women, aged 25–35, who volunteered to participate in the study. Eleven of these people were recruited by a market research firm and the twelfth was recruited by a member of the

research team. All were regular cell phone users who wanted to increase their physical activity. None of the participants knew each other. They came from a wide range of occupations, including marketing specialist, receptionist, elementary school employee, musician, copywriter, and more. Eight were full-time employed, two were homemakers, one was employed part time, and one was self-employed. Six participants were classified as overweight, five as normal, and one as obese, based on body mass calculations.

The study lasted for 21 to 25 days, during which the participants were interviewed individually three times. The first interview session focused on their attitudes to physical activity and included setting their own activity goals. In the second interview sessions (day 7) participants were allowed to revise their weekly activity schedule. The last interview session took place on day 21. These interviews were recorded and transcribed. The participants were compensated for their participation.

Data analysis and presentation.

[Figure 14.9](#) shows the data that the evaluators collected for each participant for the various exercises. Some of the data was inferred by the system, some was manually written up in a journal, and some was a combination of the two. The way in which they were recorded over time and participant varied (the participants are represented by numbers in the vertical axis and the day of the study is represented by the horizontal axis). The reason for this is that sometimes the system inferred activities incorrectly which the participants then changed. An example was housework, which was inferred as bicycling. Manually written up activities (described as 'journaled' in the figure) included such things as swimming and weightlifting, which the system could not or was not trained to record.

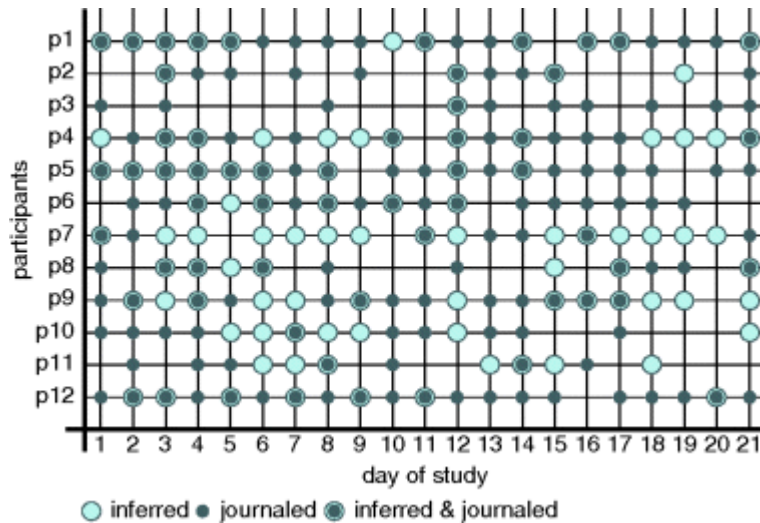


Figure 14.9 Frequency of performed activities and how they were recorded for each participant

Source: From Consolvo, S., McDonald, D.W., Toscos, T. et al (2008) “Activity sensing in the wild: a field trial of UbiFit garden”. In: *Proceedings of CHI 2008*, ACM Press, New York, p. 1802.

From the interviews, the researchers learned about the users’ reactions to the usability of UbiFit Garden, how they felt when it went wrong, and how they felt about it in general as a support for helping them to be fitter. Seven types of error with the system were reported. These included: making an error with the start time, making an error with the duration, confusing activities in various ways, failing to detect an activity, and detecting an activity when none occurred. These were coded into categories backed up by quotes taken from the participants’ discussions during the focus groups. Two examples are:

What was really funny was, um, I did, I did some, um a bunch of housework one night. And boom, boom , boom, I'm getting all these little pink flowers. Like, ooh, that was very satisfying to get those. (P9, Consolvo et al, 2008, p. 1803)

. . . it's not the end of the world, [but] it's a little disappointing when you do an activity and it [the fitness device] doesn't log it [the activity] . . . and then I think, ‘am I doing something wrong?’ (P2, Consolvo et al, 2008, p. 1803)

An example of a general comment was:

The silly flowers work, you know? . . . It's right there on your wallpaper so every time you pick up your phone you are seeing it and you're like, 'Oh, look at this. I have all those flowers. I want more flowers.' It's remarkable, for me it was remarkably like, 'Oh well, if I walk there it's just 10 minutes. I might get another flower. So, sure, I'll just walk.' (P5, Consolvo et al, 2008, p. 1804)

Overall the study showed that participants liked the system (i.e. the user experience). Some participants even commented about how the system motivated them to exercise. However, there were also technical and usability problems that needed to be improved, especially concerning the accuracy of activity recording.

Activity 14.3

1. Why was UbiFit Garden evaluated in the wild rather than in a controlled laboratory setting?
2. Two types of data are presented from the field study. What does each contribute to our understanding of the study?

Comment

Show/Hide

14.4.2 Other Perspectives

Field studies may be conducted where a behavior of interest to the researchers only reveals itself after a long time of using a tool. For example, the expected changes in user problem-solving strategies using a sophisticated visualization tool for knowledge discovery may only emerge after days or weeks of active use (Shneiderman and Plaisant, 2006). To evaluate the efficacy of such tools, users are best studied in realistic settings of their own workplaces, dealing with their own data, and setting their own agenda for extracting insights relevant to their professional goals. An initial interview conducted by

the researchers is usually carried out to ensure the participant has a problem to work on, available data, and a schedule for completion. Then the participant will get an introductory training session, followed by 2–4 weeks of novice usage, followed by 2–4 weeks of mature usage, leading to a semi-structured exit interview. Additional assistance may be provided by the researcher as needed, thereby reducing the traditional separation between researcher and participant, but this close connection enables the researcher to develop a deeper understanding of the users' struggles and successes with the tools. More data such as daily diaries, automated logs of usage, structured questionnaires, and interviews can also be used to provide a multidimensional understanding of weaknesses and strengths of the tool.

Sometimes, a particular conceptual or theoretical framework is adopted to guide how the evaluation is performed or how the data collected from the evaluation is analyzed (see [Chapter 8](#)). This enables the data to be explained at a more general level in terms of specific cognitive processes, or social practices such as learning, or conversational or linguistic interactions. For example, Activity Theory was used as a framework to analyze how a family learned to use a new TV and video system in their own home (Petersen *et al*, 2002). Another example of theory use is semiotic engineering (de Souza, 2005), which is based on semiotic theory. In semiotic engineering, the study of signs aids designers in analyzing communication between designers and users.

Video discussing the design of scalable games by Brazilian children, aided by their teachers, using concepts from semiotic engineering, at <http://youtu.be/CYTi0bXjzYU>

Typically, studies in the field, particularly those done in the wild, are useful when evaluators want to discover how new products and prototypes will be used within their intended social and physical context of use. Routines and other types of activities are analyzed as they unfold in their natural settings, describing and conceptualizing the ways artifacts are used and appropriated. Interventions by

evaluators are limited, other than the placement of the prototype or product in the setting, and questions and/or probes to discover how the system is learned, used, and adopted. In contrast, evaluations in laboratories tend to focus on usability and how users perform on predefined tasks.

With the development of a wide variety of mobile, ambient, wearable, and other kinds of systems during the past few years, evaluators have to be creative in adapting the methods that they use to meet the challenges of participants on the move and in unusual environments.

Dilemma

How many users should I include in my evaluation study?

A question students always ask is how many users do I need to include in my study? Deciding on how many to involve in a usability study is partly a logistical issue that depends on schedules, budgets, representative users, and the facilities available. Many professionals recommend that 5–12 testers is enough for many types of studies such as those conducted in controlled or partially controlled settings (Dumas and Redish, 1999), although a handful of users can provide useful feedback at early stages of a design. Others say that as soon as the same kinds of problems start being revealed and there is nothing new, it is time to stop. The more participants there are, the more representative the findings will be across the user population, but the study will also be more expensive and time-consuming, so there is a trade-off to be made.

For field studies the number of people being studied will vary, depending on what is of interest: it may be a family at home, a software team in an engineering firm, children in a playground. The problem with field studies is that they may not be representative of how other groups would act. However, the detailed findings gleaned from these studies about how participants learn to use a technology and appropriate it over time can be very revealing. ■

Assignment

This assignment continues work on the online booking facility introduced at the end of [Chapter 10](#) and continued in [Chapter 11](#). Using any of the prototypes you have developed to represent the basic structure of your product, follow the instructions below to evaluate your prototype:

- a. Based on your knowledge of the requirements for this system, develop a standard task (e.g. booking two seats for a particular performance).
- b. Consider the relationship between yourself and your participants. Do you need to use an informed consent form? If so, prepare a suitable informed consent form. Justify your decision.
- c. Select three typical users, who can be friends or colleagues, and ask them to do the task using your prototype.
- d. Note the problems that each user encounters. If you can, time their performance. (If you happen to have a camera or a smartphone with a camera, you could film each participant.)
- e. Since the system is not actually implemented, you cannot study it in typical settings of use. However, imagine that you are planning a controlled usability study or a field study. How would you do it? What kinds of things would you need to take into account? What sort of data would you collect and how would you analyze it?
- f. What are the main benefits and problems with doing a controlled study versus studying the product in a natural setting?

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Summary

This chapter described evaluation studies in different settings. It focused on controlled laboratory studies, experiments, and field studies in natural settings. A study of the iPad when it first came out was presented as an example of usability testing. This testing was done in an office environment; this was controlled to avoid outside influences affecting the testing. The participants were also asked to conduct predefined tasks that the evaluators were interested in investigating. Experimental design was then discussed that involves testing a hypothesis in a controlled research lab. The chapter ended with a discussion of field studies in which participants use new technologies in natural settings. The UbiFit Garden example involved evaluating how participants used a mobile fitness system designed to encourage people to do daily exercise. The goal of the evaluation was to examine how participants used the system in their daily lives, what kinds of problems they encountered, and whether they liked the system.

Key differences between usability testing, experiments, and field studies include the location of the study – usability or makeshift usability lab, research lab, or natural environment – and how much control is imposed. At one end of the spectrum is laboratory testing and at the other are in the wild studies. Most studies use a combination of different methods and evaluators often have to adapt their methods to cope with unusual new circumstances created by the new systems being developed.

Key points

- Usability testing takes place in usability labs or temporary makeshift labs. These labs enable evaluators to control the test setting.
- Usability testing focuses on performance measures such as how long and how many errors are made, when completing a set of predefined tasks. Indirect observation (video and

keystroke logging) is conducted and supplemented by user satisfaction questionnaires and interviews.

- Remote testing systems have been developed that are more affordable than usability labs and also more portable. Many contain mobile eye-tracking and other devices.
- Experiments aim to test a hypothesis by manipulating certain variables while keeping others constant.
- The experimenter controls independent variable(s) in order to measure dependent variable(s).
- Field studies are evaluation studies that are carried out in natural settings; they aim to discover how people interact with technology in the real world.
- Field studies that involve the deployment of prototypes or technologies in natural settings may also be referred to as 'in the wild'.
- Sometimes the findings of a field study are unexpected, especially for in the wild studies in which the aim is usually to explore how novel technologies are used by participants in their own homes, places of work, or outside.

Further Reading

BUDI, R. and NIELSEN, J. (2012) *Mobile Usability*. New Riders Press. This book asks and attempts to answer the question of how we create usability and a satisfying user experience on smartphones, tablets, and other mobile devices.

CRABTREE, A., CHAMBERLAIN, A., GRINTER, R.E., JONES, M., RODDEN, T. and ROGERS, Y. (2013) Introduction to the special issue of "The Turn to The Wild" *ACM Transactions on Computer-Human Interaction (TOCHI)*, 20 (3). This collection of articles provides in-depth case studies of projects that were conducted in the wild over many years, from the widespread uptake of children's

story-telling mobile apps to the adoption of online community technologies.

Dumas, J. S. and Redish, J. C. (1999) *A Practical Guide to Usability Testing*. Intellect. Many books have been written about usability testing, but though now old, this one is useful because it describes the process in detail and provides many examples.

KRUG, S. (2014) *Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability (3rd edn)*. Morgan Kaufmann. This revised book provides many common-sense suggestions for doing usability testing.

Lazar, J., FENG, J. and HOCHHEISER, H. (2010) *Research Methods in Human–Computer Interaction*. John Wiley & Sons Ltd. This book discusses experimental design in detail as well as other research methods such as surveys.

Robson, C. (1994, 2011) *Experimental Design and Statistics in Psychology*. Penguin Psychology. Though now old, this book provides a useful introduction to experimental design and basic statistics. Another useful book by the same author is *Real World Research* (3rd edn), published in 2011 by Blackwell Publishing.

Shneiderman, B. and PLAISANT, C. (2010) *Designing the User Interface: Strategies for effective human–computer interaction* (5th edn). Addison-Wesley. [Chapter 4](#) provides a good overview of all the main evaluation techniques used in HCI.